

FFC Sona Urea Project

AI Use and Verification Log

Oct 5, 2026

Company-Calibrated Synthetic Dataset for Academic Analysis — illustrative figures, calibrated to Fauji Fertilizer Company Limited's disclosed scale, not actual company records.

AI (Claude) was used throughout this project to read the FFC annual report, generate the synthetic dataset, write code, build the dashboard and storyboard, and audit the result. Every AI output was checked: by recomputation, by deliberate leakage tests, against the annual report, and finally by an independent audit that reversed two earlier conclusions. This log records the prompts, the checks, and the corrections.

AI tools and their roles

Tool	Used for	Not used for
Claude (chat, with code execution)	Reading the FFC Annual Integrated Report 2025; generating and freezing the synthetic dataset; Python analysis; Colab code; charts	The Session 5 error-metric calculation, which the course requires by hand or in plain Excel first
Claude Artifacts	Interactive Forecasting Dashboard v1 and the 6-frame storyboard	Inventing any figure: every number traces to the frozen dataset, the annual report or a stated assumption
Google Colab (student-run)	Re-running every model, metric and stress test from the notebook cells	n/a
Excel (student-run)	Recomputing descriptive statistics and error metrics as a manual check	n/a

Prompt log by section

Section	Prompt given to AI (summary)	AI output	How it was checked	Correction or note
Decision	Choose one FFC product and state the decision in one sentence before any data	Sona Urea; quarterly 2026 production and stock plan, commit now vs keep flexible	Product share and 2025 quarterly figures checked against the annual report (pp. 55, 77–80)	None
Data (4.2)	Course template prompt: 104–156 weekly rows, calibrated to FFC, with price, promotion, event, driver, stockout and shock columns	156-week dataset v1	Quarterly totals tied to the annual report; regression-readiness check run	v1 failed readiness (promotion coverage 10.9%, price changed every week); regenerated as v2 before any model was fitted, then frozen with a checksum
Data dictionary and assumption log	Build the dictionary and assumption log in the course structure	Dictionary, assumption log, verification checklist	Student spot-checks: quarterly sums, 26 promotion weeks, flood-week dip	Student must rewrite the assumption log in their own words
Demand profile	Plot, describe and diagnose the series before forecasting	Annotated chart, statistics by segment, 143-word diagnosis	Statistics recomputed in Excel	None

Section	Prompt given to AI (summary)	AI output	How it was checked	Correction or note
Error fundamentals (B)	Compute and interpret MAD, MAPE, RMSE, WAPE, Bias, RSFE, Tracking Signal	Formula-driven Excel workbook	Workbook recalculated and matched to Python; the Session 5 by-hand rule flagged	Student does the by-hand calculation first and notes AI was used only to check
Model comparison (8)	Write Colab code for Naive, Seasonal Naive, SMA, SES (0.3, 0.6) and a seasonal method with Event_Flag; fit on train only	Six methods, comparison table, chart	Fixed-origin forecasts; seasonal naive inputs confirmed inside training data	Winner later changed by the audit (see Corrections)
Holdout validation (9)	Mark train and holdout, find five largest misses, prove no leakage	Validation chart, misses, six leakage checks	Scramble and delete tests: forecasts unchanged	Tracking signal claim corrected by the audit
Regression (10)	Course prompt: regress demand on Price, Promo_Flag, Event_Flag and Economic_Index; report and interpret	Model A (required) and Model B (with Merger_Flag)	Variance inflation factors; sign check against business logic	Price elasticity withdrawn by the audit
Stress test (11)	Stress the decision under downside, upside, event and operational scenarios	Weekly stock simulation, 12 scenarios	Inputs traced to the annual report or labelled as assumptions	Baseline, discount cost and plan revised by the audit
Dashboard (12) and storyboard (13)	Build an executive dashboard (zones A–H) and a 6-frame storyboard	Published dashboard and slide deck	Rendered and checked on desktop, phone and dark mode; every number traced to the analysis	Updated after the audit
Recommendation (14)	State a quantified decision with a review trigger	Recommendation and Word document	Triggers calibrated against measured forecast error	A 2,721 KT figure corrected to 2,571 KT before delivery
Audit	Act as an independent forecasting auditor; do not assume the model is correct	Independent recomputation and challenge	Fresh code written from the frozen CSV	Seven corrections applied (see next section)

Corrections made to AI output

The two most important corrections came from the independent audit and changed the recommendation, not just the wording.

#	Issue found	Found by	Fix
1	Dataset v1 failed the readiness check: promotions in 10.9% of weeks; price changed every week	Readiness check	Regenerated as v2 before any model was fitted; frozen with checksum 0d23533...
2	v2 draft included a July 2025 price increase, contradicting the annual report (no inflationary impact passed on in 2025)	AI self-check against the report	Removed before freezing
3	Data tuning disclosure: while building v2, the economic-index path and the strength of the price effect were adjusted so the built-in effects could be recovered by a regression	AI disclosure	Disclosed here. The regression therefore partly reflects design choices and validates the method, not FFC's market
4	Required regression (Model A) gave price a positive sign	AI interpretation	Explained as merger confounding; Model B adds Merger_Flag

#	Issue found	Found by	Fix
5	Model choice: seasonal decomposition was selected on one 12-week holdout	Independent audit	Rolling test over 17 windows: decomposition won only 9 and failed at the 2025 turning point. Switched to the average of decomposition and seasonal naive (18.1% mean WAPE, the lowest)
6	Tracking signal reported as -3.7 , within ± 4 , but it breached -4 in weeks 5 to 10	Independent audit	Path reported; the averaged forecast stays inside ± 4 every week
7	Price elasticity of -0.4 claimed, but the coefficient changes sign across specifications	Independent audit	Elasticity withdrawn; price effect reported as not reliably identified
8	Historical_Forecast described as FFC's own planning forecast	Independent audit	Relabelled as a synthetic planner forecast
9	Merger scope factor ($\times 1.184$) uses full-year 2025 figures that include the holdout quarter; the holdout was also reused to compare model variants	Independent audit	Disclosed on the dashboard and in the report; a training-only factor ($\times 1.09$) changes holdout WAPE only from 14.6% to 14.8%
10	Discount cost per extra bag used 2023-era prices (PKR 1,250)	Independent audit	Updated to PKR 1,479 at 2026 prices
11	Plan: full-rate production assumed a 3,149 KT baseline, best-quarter capacity and an unverified opening stock	Independent audit	Central forecast lowered to 2,948 KT; plan changed to Port Qasim at 50% in Q1 with an end-Q1 trigger; capacity and opening-stock checks added

Verification methods

- **Data freeze:** dataset v2.0 frozen on 5 Oct 2026 with random seed 2026 and a SHA-256 checksum; no regeneration after model comparison began.
- **Date and split checks:** 156 consecutive Mondays with no gaps or duplicates; 144 training weeks end before the 12 holdout weeks begin.
- **Leakage tests:** holdout forecasts are identical when holdout demand is scrambled, deleted, or reduced to calendar fields only.
- **Recomputation:** every reported metric was recomputed with independent code (and in Excel by the student) and matched exactly.
- **Rolling-origin backtest:** 17 rolling 12-week windows, so model choice does not rest on one holdout.
- **Regression diagnostics:** variance inflation factors, Durbin–Watson (1.35), Newey–West standard errors, and coefficient stability across alternative specifications.
- **Source checks:** every real figure traced to the FFC Annual Integrated Report 2025; every assumption labelled as such.

Limitations. All demand data are synthetic. The regression recovers effects that were built into the data, so it demonstrates method, not FFC's market. Opening stock (150 KT) and full-rate capacity (60.8 KT a week) are assumptions that materially change the plan. The holdout covers one quarter and one Rabi season.